SECURITY OPERATIONS CENTRE (SOC) INCIDENT INVESTIGATION AND RESPONSE REPORT

Project Title: Threat Detection, Investigation and Malware Response Using Wazuh SIEM and VirusTotal

Role: Security Operations Centre (SOC) Analyst
SIEM Platform: Wazuh
Endpoint: Ubuntu 2 (Ubuntu-VirtualBox)
Threat Intelligence Platform: VirusTotal
Investigation Period: 22–29 October 2026
Incident Classification: Suspicious Privilege Escalation, Account Creation and Malware Detection

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1. Executive Summary

As a SOC Analyst, I conducted a security monitoring, threat detection and incident response exercise using Wazuh Security Information and Event Management (SIEM), integrated with VirusTotal for threat intelligence enrichment.

The primary objective was to identify suspicious activities, investigate security alerts, enrich indicators of compromise (IOCs), implement appropriate remediation measures and verify that the threats had been successfully removed.

During the investigation, Wazuh generated several security alerts involving privileged command execution, suspicious account creation and file integrity changes on the Ubuntu 2 endpoint.

A separate malware investigation identified a suspicious file, eicar.com, which was submitted to VirusTotal for analysis. The threat intelligence investigation returned a detection result from 64 antivirus security vendors, identifying the file as malicious and associating it with Trojan-related classifications and suspicious IP indicators.

Following the investigation, I configured an automated active response mechanism using a Bash script to remove the malicious file from the affected endpoint. Wazuh subsequently recorded the response, and file integrity monitoring provided evidence that the suspicious file was no longer present.

The investigation demonstrated the practical application of the SOC incident management lifecycle: Detect, Investigate, Enrich, Respond and Verify.

2. Incident Detection

SOC Stage 1: DETECT

The first stage involved monitoring security events generated by the Ubuntu endpoint through the Wazuh SIEM platform.

Wazuh collected and analysed endpoint logs, identifying potentially suspicious activities through its predefined security rules.

Three significant alerts were identified during the investigation.

2.1 Suspicious Privileged Command Execution

- **Date:** 22 October 2026

- **Time:** 12:42:52
- **Endpoint:** Ubuntu 2
- **Wazuh Rule ID:** 5502
- **Severity:** Medium
- **Command:** sudo id
- **Activity:** A user executed the id command with administrative privileges.
- **MITRE ATT&CK mapping recorded:** Defence Evasion, Impact and Privilege Escalation.

The alert indicated that the Ubuntu user account executed a command with root privileges. The command displayed the user identity, user ID (UID), group ID (GID) and group memberships.

Although privileged command execution can be legitimate administrative activity, unexpected elevation of privileges may indicate an attempt to gain unauthorised access to sensitive system resources.

2.2 Privileged Shell Execution

- **Date:** 22 October 2026
- **Time:** 13:21:53
- **Endpoint:** Ubuntu 2
- **Wazuh Rule ID:** 5502
- **Severity:** Low
- **Command:** sudo sh -c 'id;whoami;hostname'
- **Activity:** Execution of multiple system identification commands within a privileged shell.
- **MITRE ATT&CK mapping recorded:** Privilege Escalation.

The command executed three system reconnaissance activities:

1. id – Identified the current user and group privileges.
2. whoami – Identified the account executing the command.
3. hostname – Retrieved the endpoint's hostname.

The command was executed through a shell with administrative privileges, allowing the user to obtain information about the system's elevated access context.

2.3 Suspicious User Account Creation

- **Date:** 22 October 2026
- **Time:** 16:02:11
- **Endpoint:** Ubuntu 2
- **Wazuh Rule ID:** 5902
- **Severity:** Medium

- **Command:** sudo useradd -m soc_test && sudo passwd -d soc_test
- **Activity:** Creation of a new user account named soc_test and execution of a password modification command.
- **MITRE ATT&CK mapping recorded:** Persistence.

The activity generated a file integrity monitoring alert indicating a checksum change.

Creating an additional user account and modifying its password configuration can be legitimate administrative activity. However, unauthorised account creation may allow an attacker to establish persistent access to a compromised endpoint.

This alert was therefore identified as requiring further investigation.

3. Investigation and Analysis

SOC Stage 2: INVESTIGATE

Following the detection of the alerts, I investigated the activities to establish their nature, potential impact and whether they represented legitimate administrative operations or suspicious behaviour.

3.1 Privilege Escalation Investigation

I reviewed the Wazuh security events associated with Rule ID 5502.

The investigation established that the Ubuntu user account (UID 1000) successfully opened a privileged session as the root user (UID 0).

The subsequent execution of id, whoami and hostname demonstrated that the user was checking the identity and administrative context of the system.

Analyst assessment:

The activities demonstrated successful privileged command execution. Although the commands were not inherently malicious, they represented activities that required monitoring because attackers may use similar commands during privilege escalation and system reconnaissance.

3.2 Account Creation Investigation

I examined the event associated with Wazuh Rule ID 5902.

The investigation identified the creation of a new account named soc_test, followed by execution of a password modification command.

Wazuh also recorded a file integrity checksum change.

Analyst assessment:

The creation of an additional account could provide an alternative access route to the endpoint. If performed without authorisation, this activity could represent an attempt to establish persistence.

Further investigation would be required to establish whether the account was authorised, whether it was added to privileged groups and whether it was subsequently used to access the system.

3.3 Malware File Investigation

A separate investigation identified a file creation event on the Ubuntu 2 endpoint.

The recorded details were:

- **Date:** 29 October 2026
- **Time:** 11:32:50
- **Endpoint:** Ubuntu 2
- **File path:** /etc/cups/subscriptions.conf.N
- **Wazuh Rule ID:** 554
- **Event:** File added to the system.

The file activity was identified through Wazuh file integrity monitoring and required further investigation to determine whether the file was legitimate or potentially malicious.

The subsequent VirusTotal investigation identified a file named eicar.com under /root/eicar.com.

The relationship between the initial file creation event and the file submitted to VirusTotal should be confirmed through the endpoint's file hashes and event records.

4. Threat Intelligence Enrichment

SOC Stage 3: ENRICH

After identifying the suspicious file, I used VirusTotal to enrich the investigation with external threat intelligence.

This stage was important because it provided additional information about the file's reputation, security vendor detections and potential indicators of compromise.

4.1 File Hash Analysis

The following file hashes were recorded during the investigation:

SHA-256:

275a021bbfb6489e54d471899f7db9d1663fc695ec2fe2a2c4538aabf651fd0f

MD5:

44d88612fea8a8f36de82e1278abb02f

These hashes provided unique file identifiers that could be used to investigate the file without relying solely on its filename.

4.2 VirusTotal Detection Results

The file was submitted to VirusTotal for analysis.

The results indicated:

- **Detection status:** Malicious file detected.
- **Antivirus vendor detections:** 64 security engines.
- **Threat classification:** Trojan.
- **Relevant threat classification:** Exploitation for Client Execution.

- **Additional indicators:** 10 suspicious IP addresses.

VirusTotal generated an alert identifying the file as malicious.

The detection by 64 security engines provided strong supporting evidence that the file was not a legitimate application.

4.3 Threat Assessment

Based on the available threat intelligence, the file was classified as a potential Trojan.

Trojans are malicious programs that disguise their true purpose as legitimate software. Depending on their functionality, they may provide attackers with unauthorised access to systems, facilitate malicious activities or establish communication with external command-and-control infrastructure.

The presence of suspicious IP indicators further justified investigating potential external communications.

SOC Analyst Assessment:

The VirusTotal results significantly increased confidence in the malware classification and supported the decision to initiate remediation.

However, the suspicious IP addresses would require further network log analysis to establish whether the endpoint had communicated with any of them.

5. Incident Response and Remediation

SOC Stage 4: RESPOND

Following the investigation and threat intelligence enrichment, I implemented a remediation process to remove the identified malicious file from the Ubuntu endpoint.

5.1 Active Response Preparation

To enable the active response script to process JSON input, I installed the jq utility on the Ubuntu endpoint.

The following commands were executed:

sudo apt update

sudo apt -y install jq

The jq utility was required to process JSON data supplied to the active response script.

5.2 Active Response Script

I created a Bash active response script named:

/var/ossec/active-response/bin/remove-threat.sh

The purpose of the script was to automatically remove malicious files identified during security monitoring.

The active response mechanism was configured to work with Wazuh so that identified threats could be addressed through an automated remediation process.

5.3 Malware Removal

The active response mechanism was executed against the identified malicious file.

The investigation documented the following outcomes:

- The malicious file was successfully removed.
- Wazuh recorded the response action.
- The endpoint underwent further file integrity monitoring.
- The suspicious file activity was no longer observed.

SOC Analyst Assessment:

The remediation process successfully addressed the identified file-based threat according to the recorded results.

Automating the response reduced the need for manual intervention and demonstrated how SIEM platforms can support faster incident containment and remediation.

6. Incident Verification

SOC Stage 5: VERIFY

The final stage involved confirming that the remediation process had been successful and that the identified malicious file was no longer present on the endpoint.

I reviewed the subsequent Wazuh file integrity monitoring events to establish whether the suspicious activity had stopped.

The verification results indicated that:

1. The active response script successfully removed the malicious file.
2. Wazuh recorded the remediation activity.
3. Subsequent file integrity monitoring showed that the suspicious file was no longer present.
4. The previously identified suspicious file activity was no longer observed.

Verification Outcome:

The available evidence confirmed successful removal of the identified malicious file.

However, complete incident closure would also require confirming that no additional malicious files remained, checking for suspicious outbound network connections and reviewing whether the endpoint had experienced any unauthorised access.

7. Overall Incident Assessment

The investigation identified three categories of security activity:

Security activity	Severity	Assessment
Privileged command execution	Medium	Successful root-level command execution identified.
Privileged shell execution	Low	System identity and hostname reconnaissance identified.

New user account creation	Medium	Potential persistence activity requiring authorisation checks.
Malicious file detection	Malicious	VirusTotal identified 64 security vendor detections.
Malware remediation	Completed	Active response successfully removed the identified file.

Impact Assessment

The investigation demonstrated potential security risks involving:

- Unauthorised privilege escalation.
- Creation of additional user accounts.
- Potential persistence mechanisms.
- Introduction of malicious files.
- Potential communication with suspicious external IP addresses.

The available evidence confirmed malware detection and file removal. However, it did not establish whether the endpoint had been successfully compromised by an external attacker or whether data had been accessed or exfiltrated.

8. Recommendations and Preventive Measures

Based on the investigation, I recommend the following security improvements:

1. Strengthen Privileged Access Management

Review sudo permissions and restrict administrative privileges to authorised users. Monitor unexpected root-level command execution.

2. Monitor User Account Creation

Configure alerts for new account creation, password changes and modifications to privileged group memberships.

3. Improve File Integrity Monitoring

Maintain Wazuh file integrity monitoring across sensitive system directories to identify suspicious file creation, modification and deletion activities.

4. Enhance Threat Intelligence Integration

Continue integrating VirusTotal with Wazuh to enrich suspicious file alerts with external threat intelligence and improve malware identification.

5. Automate Incident Response

Develop and test additional Wazuh active response scripts to support automated malware removal and incident containment.

6. Investigate Network Indicators

Review network traffic and endpoint logs for connections to the 10 suspicious IP addresses identified during the VirusTotal investigation.

7. Conduct Post-Incident Monitoring

Continue monitoring the Ubuntu endpoint for recurring alerts, unauthorised account activity and additional indicators of compromise.

9. Conclusion

This SOC capstone project demonstrated the practical application of the Security Operations Centre incident management lifecycle using Wazuh SIEM and VirusTotal.

As the SOC Analyst, I successfully demonstrated the five essential stages of security incident handling:

- **Detect:** Identified suspicious privileged command execution, account creation and file activity through Wazuh security alerts.
- **Investigate:** Examined the alerts, reviewed endpoint activity and assessed potential privilege escalation and persistence risks.
- **Enrich:** Used VirusTotal to analyse file hashes and obtain threat intelligence, identifying 64 security vendor detections.
- **Respond:** Configured and executed a Wazuh active response mechanism to remove the identified malicious file.
- **Verify:** Reviewed subsequent file integrity monitoring events and confirmed that the suspicious file was no longer present.

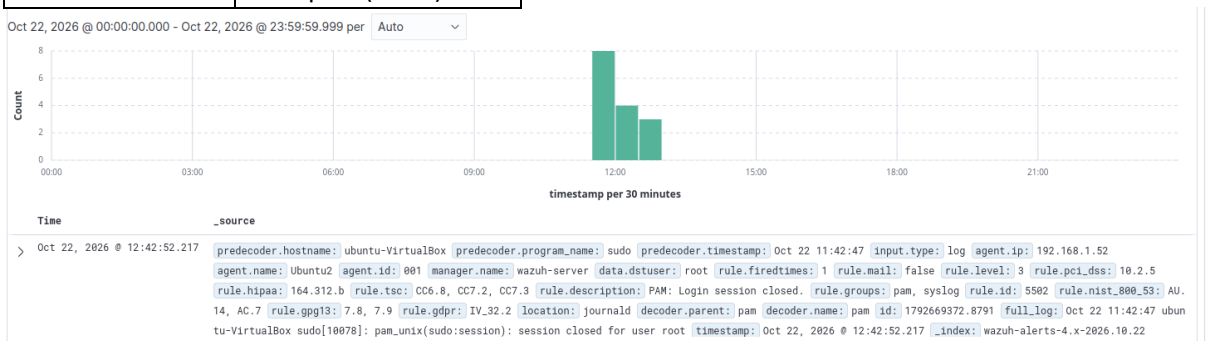
Overall, the project demonstrated how SIEM technology, endpoint monitoring, threat intelligence and automated response mechanisms can work together to improve an organisation's security monitoring and incident response capabilities.

The exercise strengthened my practical understanding of security alert triage, log analysis, threat intelligence enrichment, malware investigation, incident remediation and post-incident verification.

Final SOC Assessment: The identified malicious file was successfully removed, and the available Wazuh evidence supported successful remediation. Further endpoint and network investigations would be required before confirming that the wider environment was completely free from compromise.

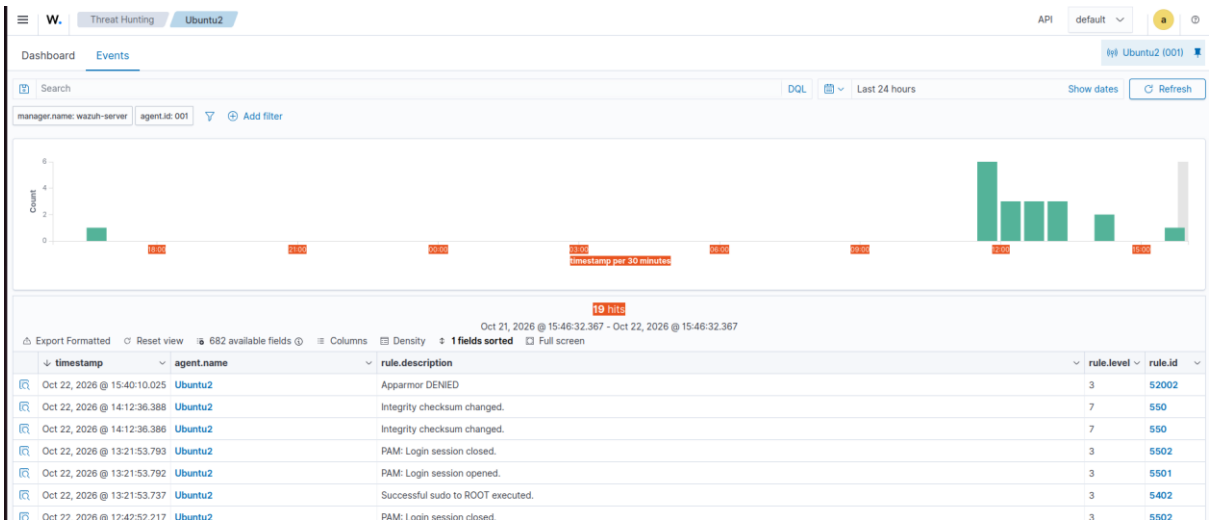
Part 1: Alert Detection & Investigation 1st Command

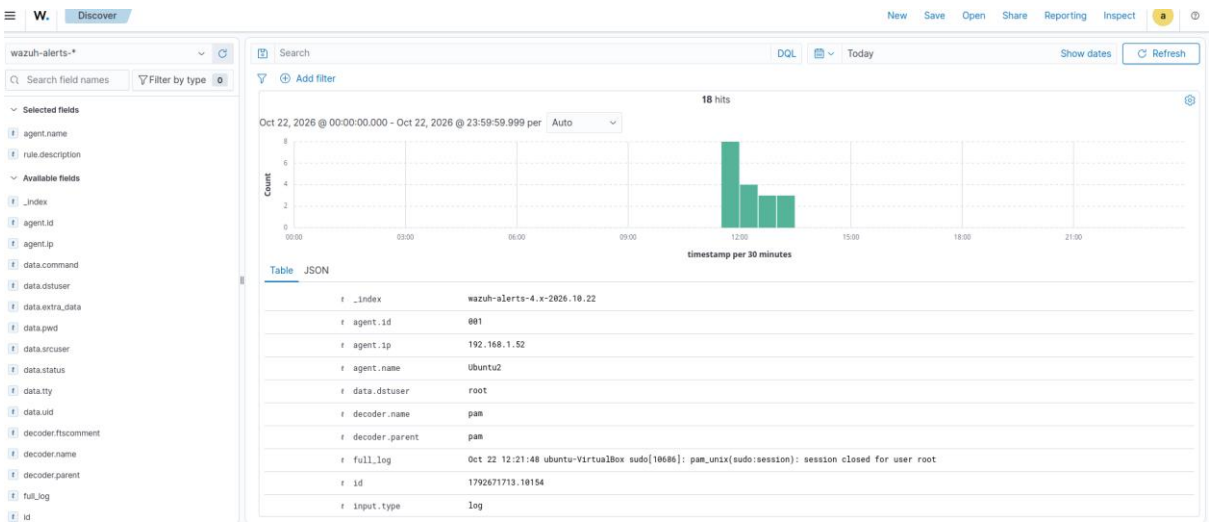
FIELD	ACTIVITY PERFORMED
sudo ID	It runs the id utility with administrative(root) privileges, displaying the user ID (UID), Group ID (GID0, and group memberships of the superuser(root, which is typically 0)
Time stamp	Oct 22, 2026 @ 12:42:52.217
Wazuh Rule	5502
Severity	Medium
Agent	Ubuntu 2
Command/Event	Oct 22 11:42:47 ubuntu-VirtualBox sudo[10078]: pam_unix(sudo:session): session opened for user root(uid=0) by ubuntu(uid=1000)
MITRE ATTACK	Defense Evasion. Impact, priviledge escalation
FInding	sudo stands for “super user do “ that temporarily elevates user’s right to run a command as an administrator, while super id prints out user identity details, including the user ID(UID0 and Group ID (GUID)

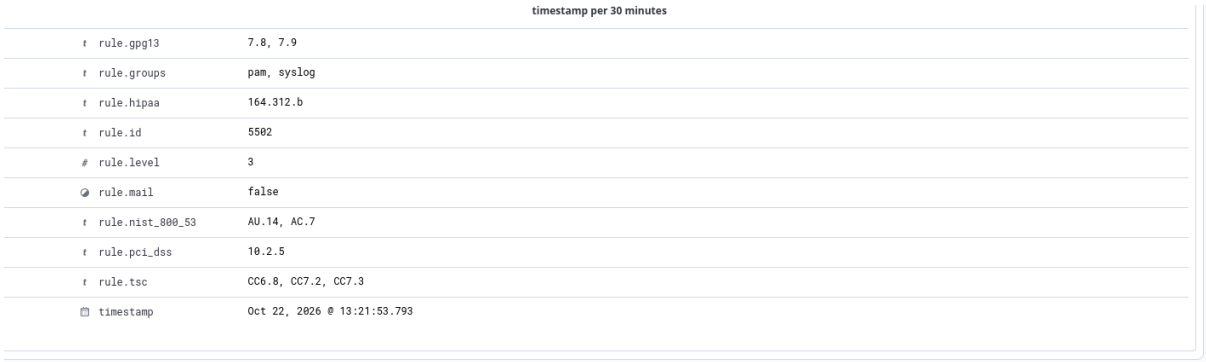


2nd command

File	Activity performed
Sudo sh -c ‘id;whoami;hostname	This series of commands allow you to execute series of systems check. This executes who am I command inside a new shell (sh) with full administrative privilege. It will output root and print hostname
Time stamp	Oct 22, 2026 @ 13:21:53.793
Wazuh Rule	5502, PAM: Login session opened.
Severity	Low
Agent	Ubuntu2
Command/Event	Oct 22 12:21:48 ubuntu-VirtualBox sudo[10686]: pam_unix(sudo:session): session opened for user root(uid=0) by ubuntu(uid=1000)
MITRE ATTACK	Priviledge esclation
Finding	This command prints the hostname and ouput the roots

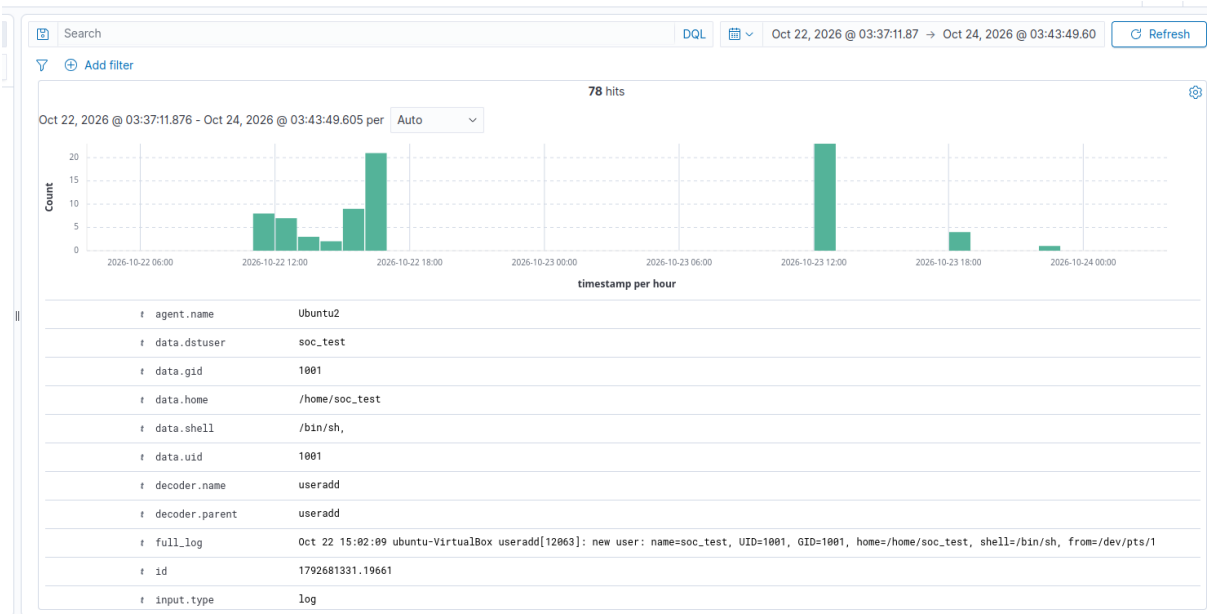






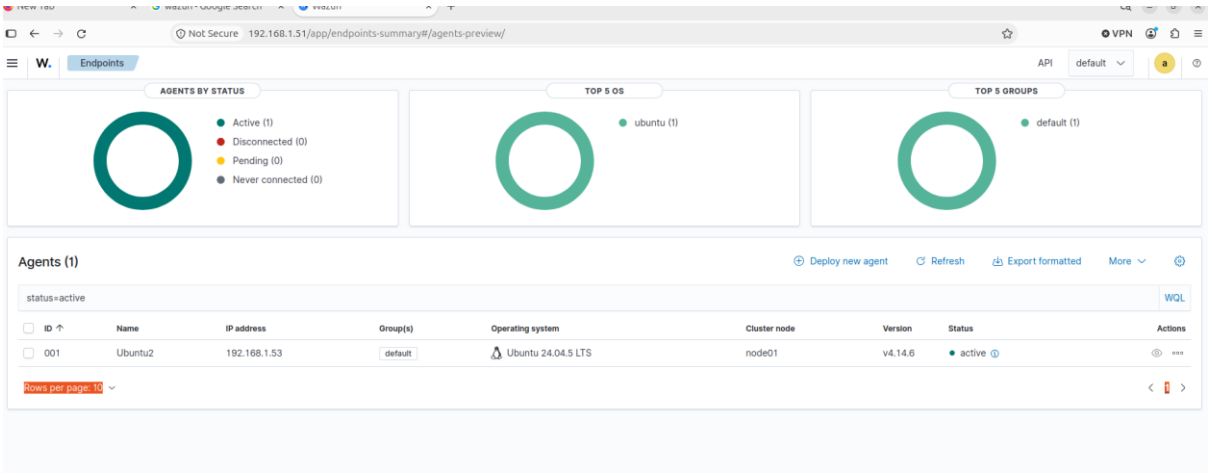
3rd command

FIELD	ACTIVITY PERFORMED
sudo useradd -m soc_test && sudo passwd -d soc_test	The command added a new user and password to the system
Time stamp	Oct 22, 2026 @ 16:02:11.642
Wazuh Rule	5902
Severity	Medium
agent	ubuntu2
User	ubuntu-VirtualBox
Command Event	Integrity checksum changed.
MITRE ATTACK	persistence
FINDING	New user called soc test was added to the group

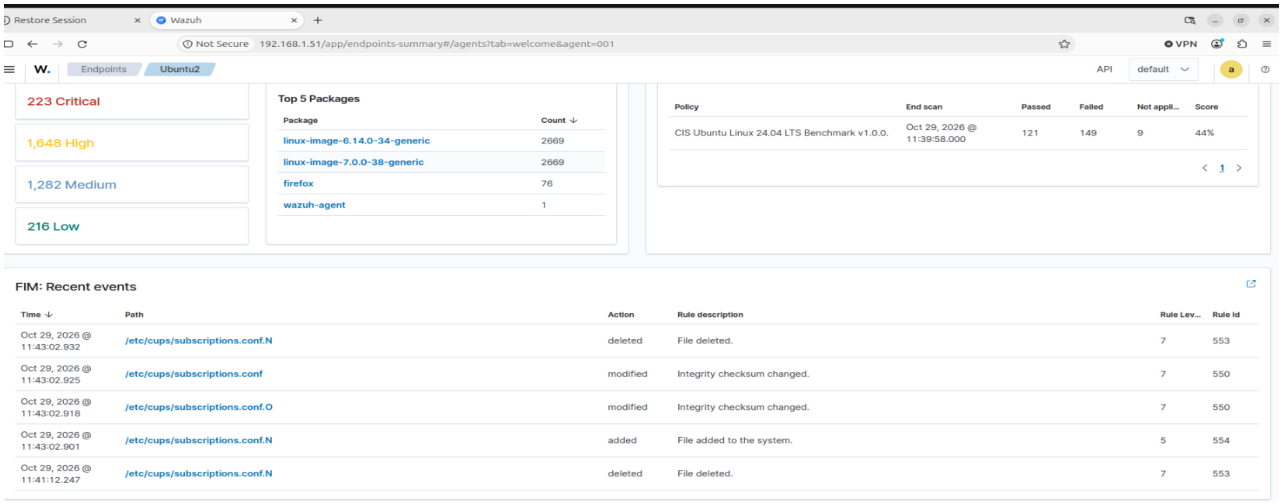


PART 2 Malware Detection & Response with Virus Total

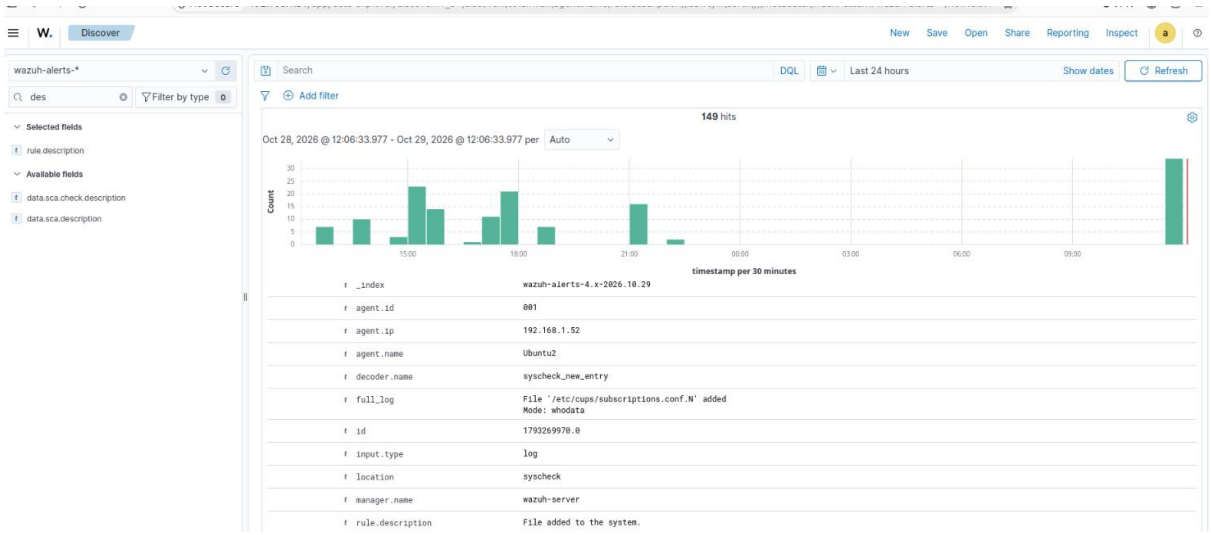
1



2



3 A file was added to the system and the path is /etc/cups/subscriptions.conf.N and ubuntu2 was the endpoint that generated the event at Oct 29, 2026 @ 11:32:50.229 and the rule id is 554



4 Enrich the investigation with virus total

File hash sha256 275a021bbfb6489e54d471899f7db9d1663fc695ec2fe2a2c4538aabf651fd0f

MDD is 44d88612fea8a8f36de82e1278abb02f

Virus total detection result VirusTotal: Alert - /root/eicar.com - 64 engines detected this file

Number of vendors detecting the file was 64

Relevant threat classification: Exploitation for client execution

Additional Indicator of compromise is that the file contains malware Trojan and 10 suspicious IPs

5. Based on the evidence the file contains a malicious malware called Trojan. This is a that misleads users as to its true intent by disguising itself as a normal program. Trojans are generally spread by some form of **social engineering.** Although their payload can be anything, many modern forms act as a backdoor, contacting a controller who can then have unauthorized access to the affected device

jq was installed, a utility that processes JSON input from the active response script.

6 jq was installed, a utility that processes JSON input from the active response script.

sudo apt update

sudo apt -y install jq

/var/ossec/active-response/bin/remove-threat.sh active response script to remove malicious files from the endpoint was created.

the action was taken so that jq can process JSON input from the active response script.

The file was successfully removed and wazuh recorded the response.

